NARUTATSU (EDWARD) RI

ASPIRING MACHINE LEARNING RESEARCHER
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ABOUT

Sophomore at Columbia Engineering interested in Machine Learning and Mathematics.

SKILLS

Proficient in Python, C/C++, MATLAB, Linux/Unix Systems
Can also use Java, SQL, HTML, CSS.

Proficient in English, Japanese, and Chinese at native level.
Elementary proficiency in Russian.

LEADERSHIP & ACTIVITIES

SUBCONTRACTOR • DATAFLUCT, INC. • 2021~
Software Engineer and Contractor at DATAFLUCT, Inc.
Currently working on an interactive business intelligence provider, responsible for the backend processing and visualization generations.
Working in a team with experienced software engineers.

RASC-AL TEAM • COLUMBIA SPACE INITIATIVE • 2020~2021
Member of CSI's RASC-AL team.
Developed plan for Venus flyby mission, and designed propulsion systems based on previous work.

SOFTWARE TEAM • COLUMBIA UNIVERSITY FORMULA RACING • 2020~2021
Member of Software team and leader of optimization group.
Conducted tire optimization and analysis with MATLAB utilizing Pacejka's Magic Formula.

DATA ANALYST • CORONAVIRUS VISUALIZATION TEAM • 2020~2020
Member of Mental Health team and conducted data scraping, analysis, and visualization.
Scraped and generated visualizations of relationships between quarantine and depression/anxiety cases. Generated map for available healthcare facilities around the United States.

CAPTAIN • FIRST ROBOTICS COMPETITION TEAM 5701 • 2017~2019
Captain of first Japanese robotics team to participate in FIRST Robotics Competition (FRC).
Raised $30,000 and restored team. Assisted in founding multiple Japanese FRC teams, partnered with FIRST Japan Headquarters and exhibited at Maker Faire Tokyo.

CAPTAIN • ROBOTICS CLUB • 2016~2019
Founder and Captain of the Robotics Club at Seiko Gakuin High School.

Convinced Principal to start Robotics Club. Gathered more than 30 students and taught programming in Python, basic electrical engineering and robotics. Awarded best booth award at school festival.

RESEARCH

INDEPENDENT RESEARCHER • 2021~ • COMPUTER SCIENCE@COLUMIBA
One-on-one research experience with Professor Nakul Verma (Columbia University).
Currently working on developing a new method for word embeddings. Conducting mathematical analysis on Skip-Gram model. Involves weekly one-on-one meetings with professor, reading multiple mathematically rich papers on a meeting basis, and validating potential new methods. Previously worked on unsupervised sentiment analysis on stock articles and devised new bootstrapping method for accuracy improvement.

SPECIAL RESEARCH FELLOW • 2017~2019 • ISYSLAB
Dialogue system R&D at iSYSLab Robot Learning Group (University of Electro-Communications).
First high school participant to publish and present research at The 33rd Annual Conference of the Japanese Society for Artificial Intelligence, 2019 (JSAI 2019): "A Dialogue System Implemented with Latent Parameters".
Approached professor through email with research idea regarding dialogue systems in 10th grade and passed special entrance laboratory examination. Conducted research for 2.5 years under Dr. Takayuki Nagai (currently Professor at University of Osaka).

HONORS & AWARDS

EGLESTON SCHOLAR • 2020 • COLUMBIA ENGINEERING
Selected as one of nine students from Columbia Engineering's class of 2024.
https://www.engineering.columbia.edu/egleston-scholars/weida-edward-li

FULL RIDE SCHOLARSHIP • 2019 • EZOE MEMORIAL FOUNDATION
49th year Academic Scholar. One of the few undergraduate students in Japan awarded a full-ride scholarship (annual, $92880).
https://www.recruit-foundation.org/academic/li/

SECOND PLACE • 2017 • PC KOSHIEN (JAPANESE NATIONAL PROGRAMMING CONTEST)
Won second place in largest national competitive programming competition.

WINNER • 2017 • ROBOCUP JAPAN OPEN (UNIVERSITY ROBOTICS COMPETITION)
Built deep learning model for translating natural language to robot commands.

FINALIST • 2018 • S-ISEF (RESEARCH ON SPACE EXPLORATION)
Presented swarm technology-based idea to officials from RIKEN and Astronauts.

THIRD PLACE • 2019 • PEAKERS MACHINE LEARNING HACKATHON
Placed 3rd in a university student-oriented competition during senior year in high school.

EDUCATION

BACHELOR'S DEGREE • 2020~2024 • COLUMBIA ENGINEERING
Egleston Scholar (Selected as top 1% of applicants within Columbia Engineering students).
Member of Columbia Undergraduate Mathematics Society and Columbia Theoretical Computer Science Group.

BACHELOR'S DEGREE • 2020~2020 • TSUKUBA UNIVERSITY
Obtained full marks on mathematics section. Attended for six months before matriculating to Columbia Engineering.

HIGH SCHOOL • 2018~2020 • SEIKO GAKUIN HIGH SCHOOL
Valedictorian of Science Division. Graduated with 4.0 GPA.

MIDDLE SCHOOL • 2014~2017 • SEIKO GAKUIN MIDDLE SCHOOL
Seiko Gakuin Middle School (聖光学院中学校高等学校) is known to be one of the most prestigious educational institutions in Japan. Students are required to pass an entrance exam in the sixth grade to gain admission.